

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description:	Trehalose dihydrate
Cat No. :	S60313
Synonyms	alpha-D-Glucopyranosyl-alpha-D-glucopyranoside
CAS No	6138-23-4
Molecular Formula	C ₁₂ H ₂₂ O ₁₁ · 2 H ₂ O

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

Details of the supplier of the safety data sheet

Importer	Supplier
Fisher Scientific Korea	Thermo Fisher Scientific Chemicals, Inc.
D5,D6, Incheon Airport Logistics Complex	30 Bond Street
150, Gonghangdong-Ro 296 Beon-Gil	Ward Hill, MA 01835-8099
Jung-Gu, Incheon	
Tel: +82-1661-9555	
Fax: +82-2-2023-0603	

E-mail address	Chem.KR@thermofisher.com
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Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture
Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements

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Other Hazards

May form explosible dust-air mixture if dispersed
This product does not contain any known or suspected endocrine disruptors

NFPA

Health
-

Flammability
1

Instability
-

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	Common Name	CAS No	Index No	Weight %
Trehalose dihydrate	alpha-D-Glucopyranosyl-alpha-D-glucopyranoside	6138-23-4	Not listed	99 - 100
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	Alpha-D-Glucopyranosyl-Alpha-D-Glucopyranoside.	99-20-7	2004-3-2874	-

SECTION 4: FIRST AID MEASURES

Description of first aid measures

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

Ingestion

Do NOT induce vomiting. Get medical attention.

Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

Self-Protection of the First Aider

No special precautions required.

Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

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No information available.

Special hazards arising from the substance or mixture

Fine dust dispersed in air may ignite.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂).

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
Trehalose dihydrate	6138-23-4	Not listed	Not listed	Not listed
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not listed	Not listed	Not listed

Component	CAS No	European Union	The United Kingdom	Germany
Trehalose dihydrate	6138-23-4	Not listed	Not listed	Not listed
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not listed	Not listed	Not listed

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ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
Trehalose dihydrate	6138-23-4	Not listed
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not listed

Exposure Controls

Engineering Measures

None under normal use conditions.

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles)

Hand Protection

Protective gloves

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment

Use only those certified by the Korea Occupational Safety and Health Administration.

Respiratory Protection

No protective equipment is needed under normal use conditions

Recommended Filter type:

Particle filter

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Off-white Powder Solid

Odor

Odorless

Odor Threshold

No data available

pH

4.5-6.5

10% aq. sol

Melting Point/Range

97 - 99 °C / 206.6 - 210.2 °F

Softening Point

No data available

Boiling Point/Range

No information available

Flash Point

No information available

Method - No information available

Evaporation Rate

Not applicable

Solid

Flammability (solid,gas)

No information available

Explosion Limits

No data available

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Vapor Pressure	No data available	Solid
Vapor Density	Not applicable	
Specific Gravity / Density	No data available	
Bulk Density	No data available	
Water Solubility	68.9 g/100ml (20°C)	
Solubility in other solvents	No information available	

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
Trehalose dihydrate	6138-23-4	No data available
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	0.3

Autoignition Temperature	No data available	Solid
Decomposition Temperature	No data available	
Viscosity	Not applicable	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Molecular Formula	C12 H22 O11 . 2 H2 O
Molecular Weight	378.32

SECTION 10: STABILITY AND REACTIVITY

<u>Reactivity</u>	None known, based on information available
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<u>Chemical Stability</u>	Stable under normal conditions.
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Possibility of Hazardous Reactions

Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

<u>Conditions to Avoid</u>	Incompatible products. Excess heat. Avoid dust formation.
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<u>Incompatible Materials</u>	Strong oxidizing agents.
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<u>Hazardous Decomposition Products</u>	Carbon monoxide (CO). Carbon dioxide (CO ₂).
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SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure

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Inhalation	May cause irritation of respiratory tract. May be harmful if inhaled.
Ingestion	May be harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.
Eyes	Irritating to eyes.
Skin	Irritating to skin. May be harmful in contact with skin.

Information on Health Hazards

(a) acute toxicity;

Oral	Based on available data, the classification criteria are not met
Dermal	No data available
Inhalation	No data available

Toxicology data for the components

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
Trehalose dihydrate	6138-23-4	4600 mg/kg (Rat)	No data available	No data available
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	4600 mg/kg (Rat)	No data available	No data available

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory	No data available
Skin	No data available

Component	CAS No	Test method	Test species	Study result
Trehalose dihydrate	6138-23-4	No data available	No data available	No data available
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	No data available	No data available	No data available

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
Trehalose dihydrate	6138-23-4	No data available	No data available	No data available
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	No data available	No data available	No data available

(f) carcinogenicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Trehalose dihydrate	6138-23-4	No data available	No data available	No data available
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	No data available	No data available	No data available

There are no known carcinogenic chemicals in this product

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
Trehalose dihydrate	6138-23-4	Not listed	Not listed	Not listed	Not listed	Not listed
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not listed	Not listed	Not listed	Not listed	Not listed

(g) reproductive toxicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Trehalose dihydrate	6138-23-4	No data available	No data available	No data available

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.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	No data available	No data available	No data available
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(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable
Solid

Other Adverse Effects

No information available.

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains.

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Trehalose dihydrate	6138-23-4	No data available	No data available	No data available	No data available
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	No data available	No data available	No data available	No data available

Persistence and degradability

Persistence Soluble in water, Persistence is unlikely, based on information available.

Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	0.3	No data available

Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
Trehalose dihydrate	6138-23-4	Not listed
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not listed

Other adverse effects No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and

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national hazardous waste regulations to ensure complete and accurate classification.

Contaminated Packaging

Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

Not Regulated

IATA

Not regulated

IMDG/IMO

Marine Pollutant

Not regulated

No hazards identified

Special Precautions for User

No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
Trehalose dihydrate	6138-23-4	-	-	-	X	-	-	X	-	X	X
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	2004-3-2874	X	202-739-6	X	X	-	-	X	X	X

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Annex 2 - 2004-3-2874	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

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Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
Trehalose dihydrate	6138-23-4	Not listed	Not listed
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not listed	Not listed

US Management Information

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OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable

CERCLA

Not applicable

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable
.alpha.-D-Glucopyranoside, .alpha.-D-glucopyranosyl	99-20-7	Not applicable	Not applicable	Not applicable

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Based on available data, the classification criteria are not met.

May form combustible dust concentrations in air.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Prepared By

Creation Date

Revision Date

Revision Number

Health, Safety and Environmental Department

29-Oct-2010

12-Jun-2024

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Revision Summary

New emergency telephone response service provider.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet